US012416471B2

(12) **United States Patent**
Schuda

(10) **Patent No.: US 12,416,471 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **MODULAR HOLSTER ASSEMBLY**

(71) Applicant: **Douglas Schuda**, Westerville, OH (US)

(72) Inventor: **Douglas Schuda**, Westerville, OH (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 569 days.

(21) Appl. No.: **17/815,676**

(22) Filed: **Jul. 28, 2022**

(65) **Prior Publication Data**

US 2023/0108400 A1 Apr. 6, 2023

Related U.S. Application Data

(60) Provisional application No. 63/262,047, filed on Oct. 4, 2021.

(51) **Int. Cl.**
F41C 33/04 (2006.01)

(52) **U.S. Cl.**
CPC ***F41C 33/041*** (2013.01)

(58) **Field of Classification Search**
CPC F41C 33/045; F41C 33/041; F41C 33/043; A45F 5/02; A45F 2005/026; A45F 5/021; A45F 2005/025
USPC 224/197–200
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,206,257 B1 * 3/2001 Peele A45F 5/021 224/272
8,517,234 B2 * 8/2013 Kincaid F41C 33/045 224/198
9,301,595 B2 4/2016 Tedder et al.
9,366,383 B2 * 6/2016 Chien F16M 11/16
9,696,113 B2 7/2017 Tedder et al.
10,119,783 B2 11/2018 Tedder et al.
10,274,284 B2 4/2019 Tedder et al.
10,274,296 B2 4/2019 Tedder et al.
10,712,127 B2 * 7/2020 Tedder A45F 5/021
10,883,795 B2 * 1/2021 Rogers F41C 33/043
11,035,648 B2 * 6/2021 Zhang F42B 39/02
11,439,210 B2 * 9/2022 Darby A45C 13/18
2007/0023468 A1 * 2/2007 Ford F41C 33/045 224/198
2015/0323285 A1 * 11/2015 McKendrick F41C 33/045 224/198
2016/0216065 A1 7/2016 Tedder et al.

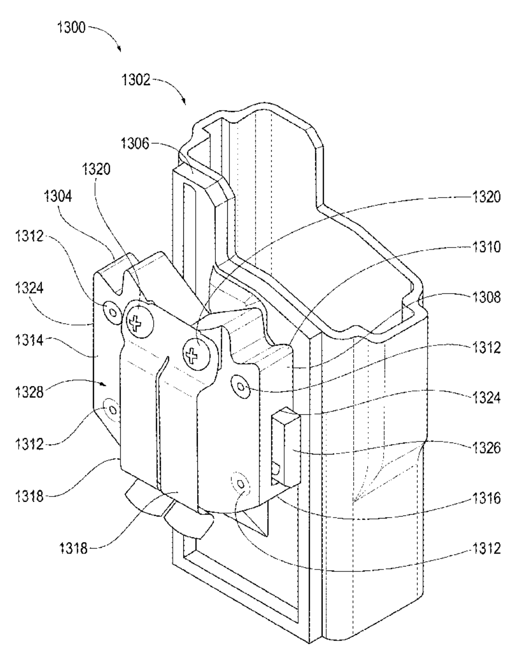
* cited by examiner

Primary Examiner — Corey N Skurdal
(74) *Attorney, Agent, or Firm* — Vorys, Sater, Seymour and Pease LLP; Rex W. Miller, II

(57) **ABSTRACT**

In an example, a mounting apparatus for releasably connecting a holster can include at least one clip and a canal. The canal can include a grooved path that can extend along a base of the canal, and an inner channel that can receive a plunger of a plunger device to secure the holster to mounting apparatus. The mounting apparatus can include a sliding pusher that can disengage the holster from the mounting apparatus. In some examples, a system can include a holster that can include a ring support structure having a cavity that can include a ring attachment structure. A mounting apparatus of the system can include a cavity that includes the slide lock, and an opening. The mounting apparatus further comprising openings through which portions of pinch plates can extend for disengaging the holster from the mounting apparatus.

5 Claims, 13 Drawing Sheets



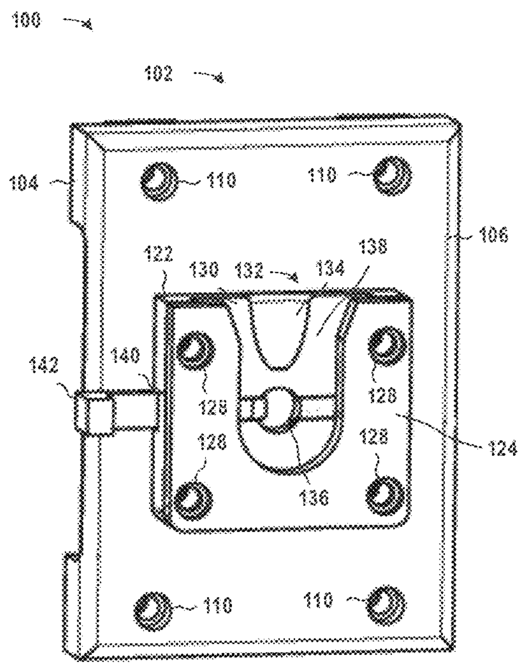

100
102
104
110
110
106
122
130
132
134
138
142
140
128
128
128
128
124
136
110
110

FIG. 1

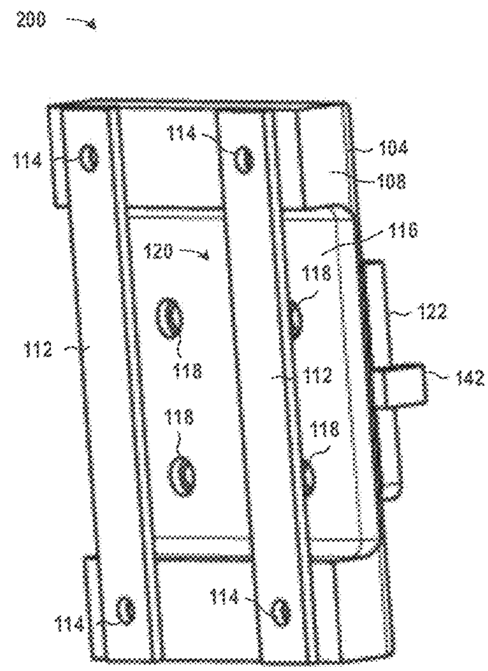

200
114
114
104
108
116
120
118
122
112
118
112
142
118
118
114
114

FIG. 2

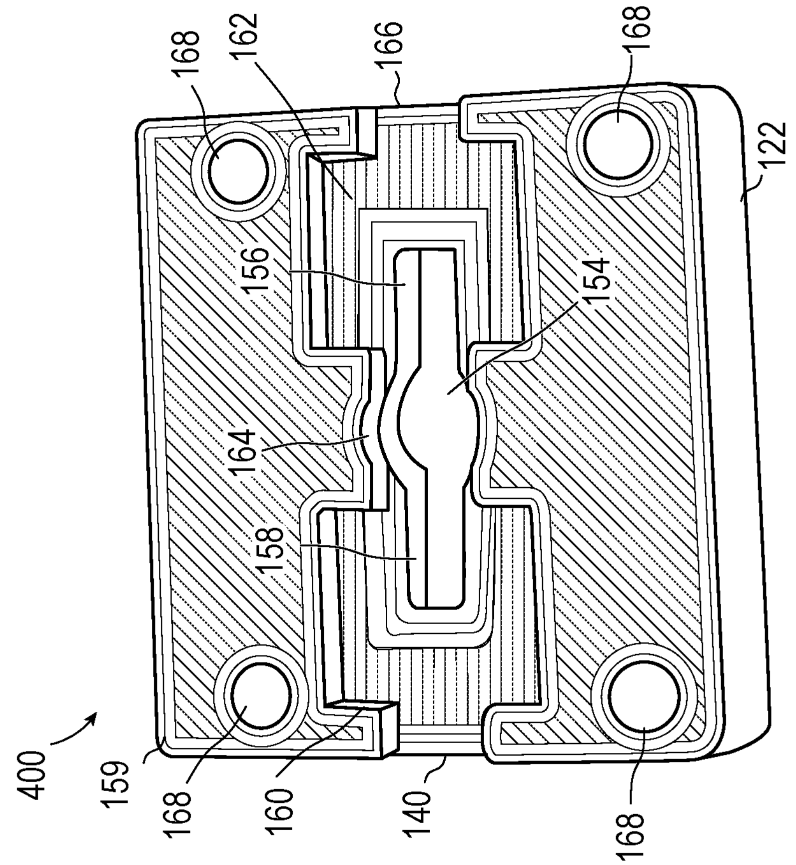

400
159
168
160
140
168
158
164
156
162
166
168
154
122
168

FIG. 4

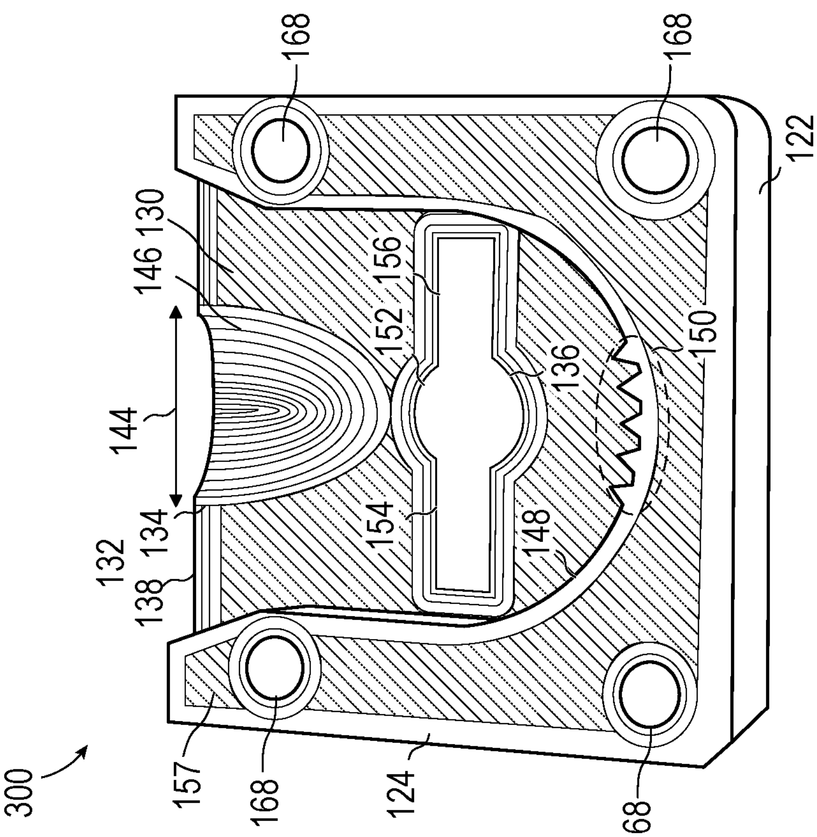

300
157
168
124
168
138
132
134
144
146
130
154
152
156
148
136
150
168
168
122

FIG. 3

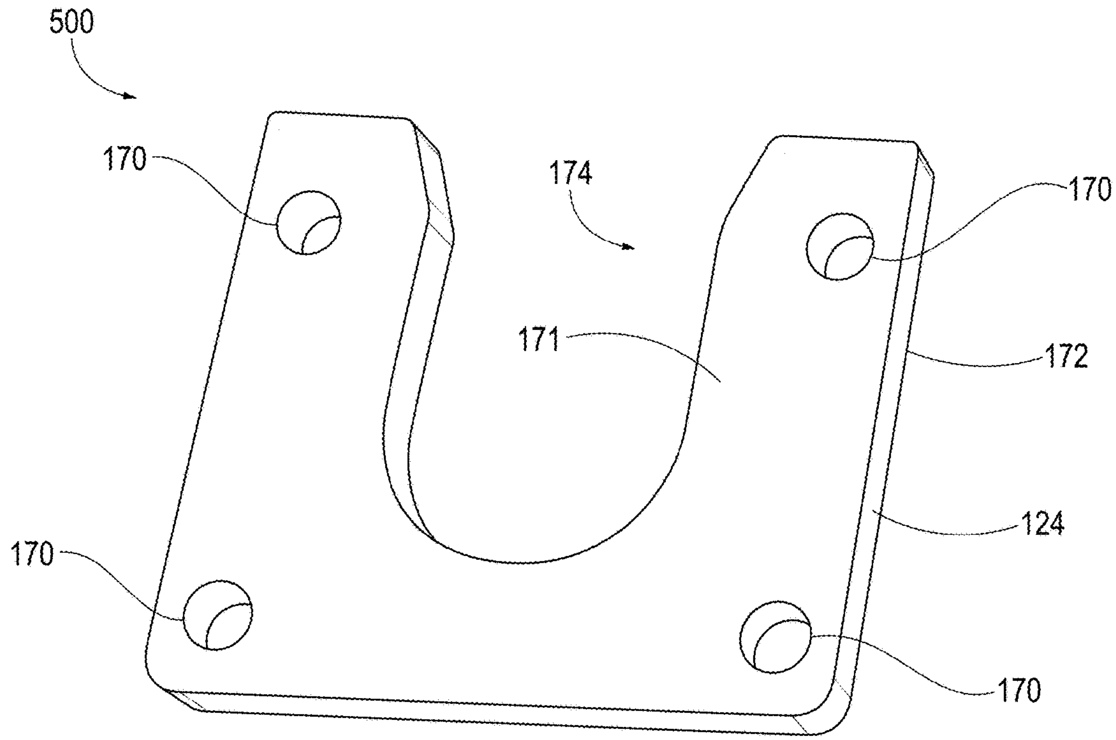


FIG. 5

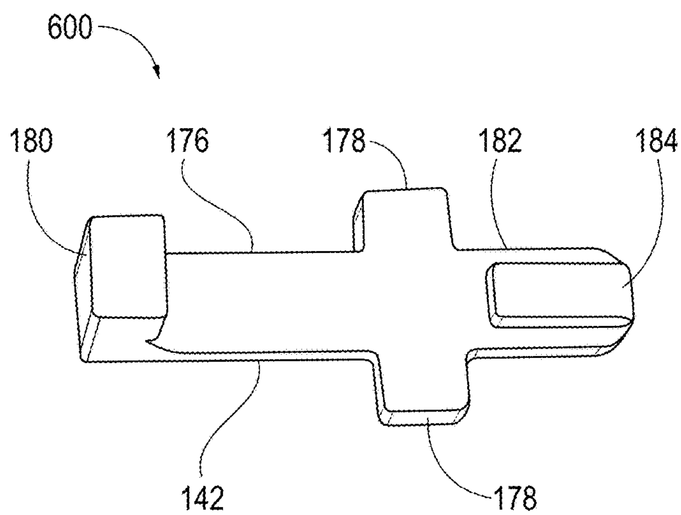


FIG. 6

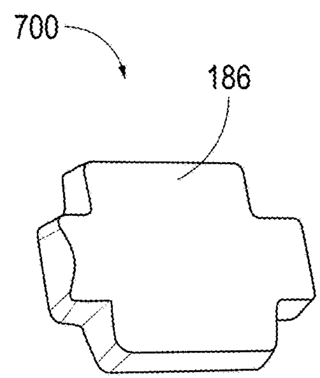


FIG. 7

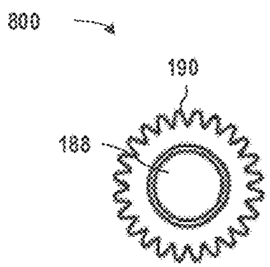


FIG. 8

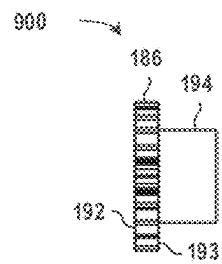


FIG. 9

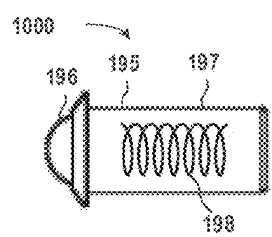


FIG. 10

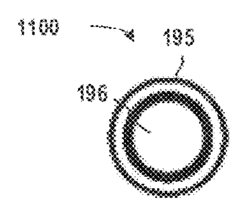


FIG. 11

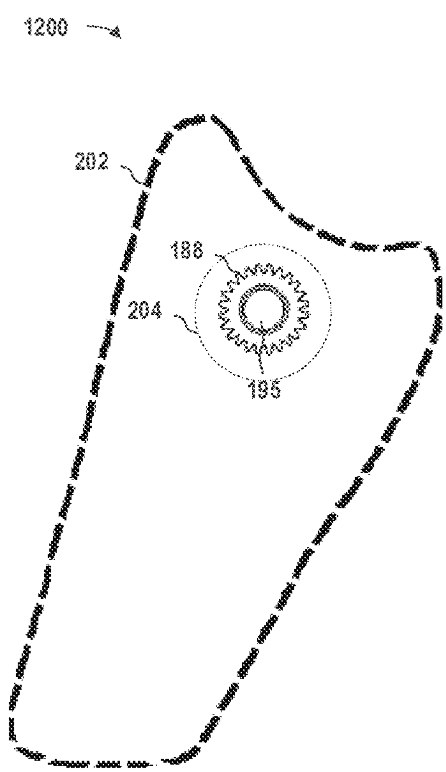


FIG. 12

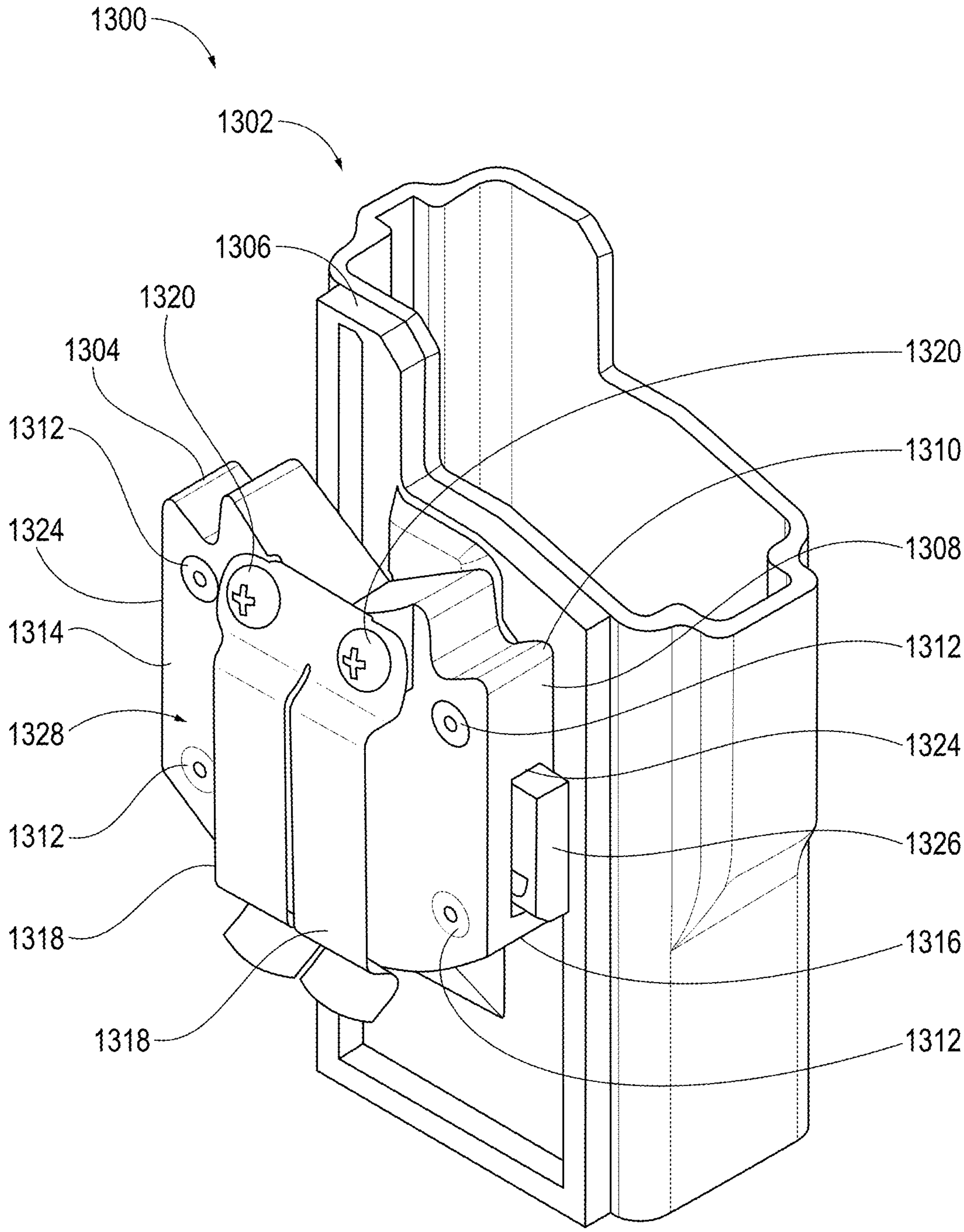

1300
1302
1306
1320
1304
1312
1324
1314
1328
1312
1318
1318
1320
1310
1308
1312
1324
1326
1316
1312

FIG. 13

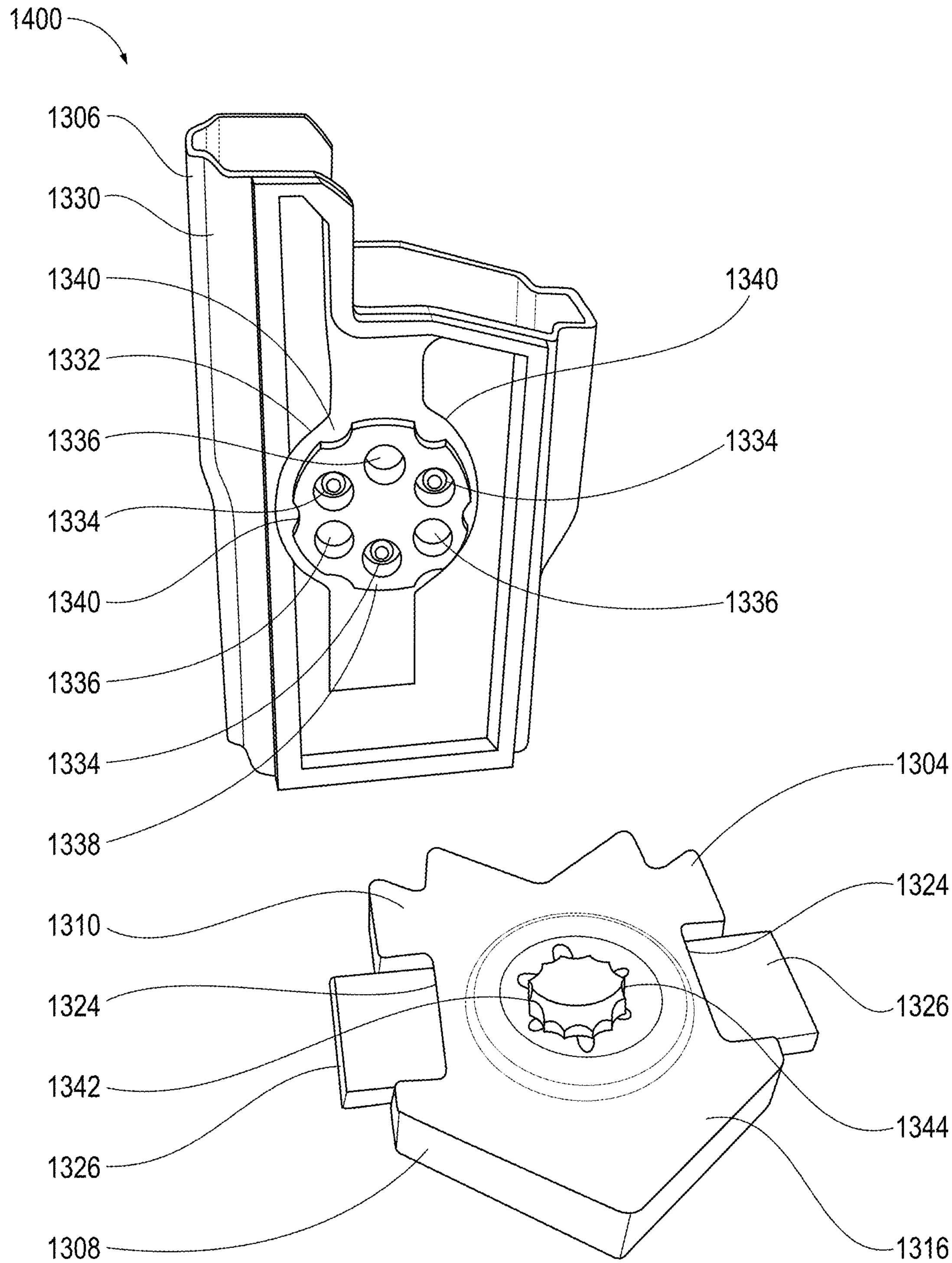

1400
1306
1330
1340
1332
1336
1334
1340
1336
1334
1338
1340
1334
1336
1304
1324
1310
1324
1326
1342
1326
1344
1308
1316

FIG. 14

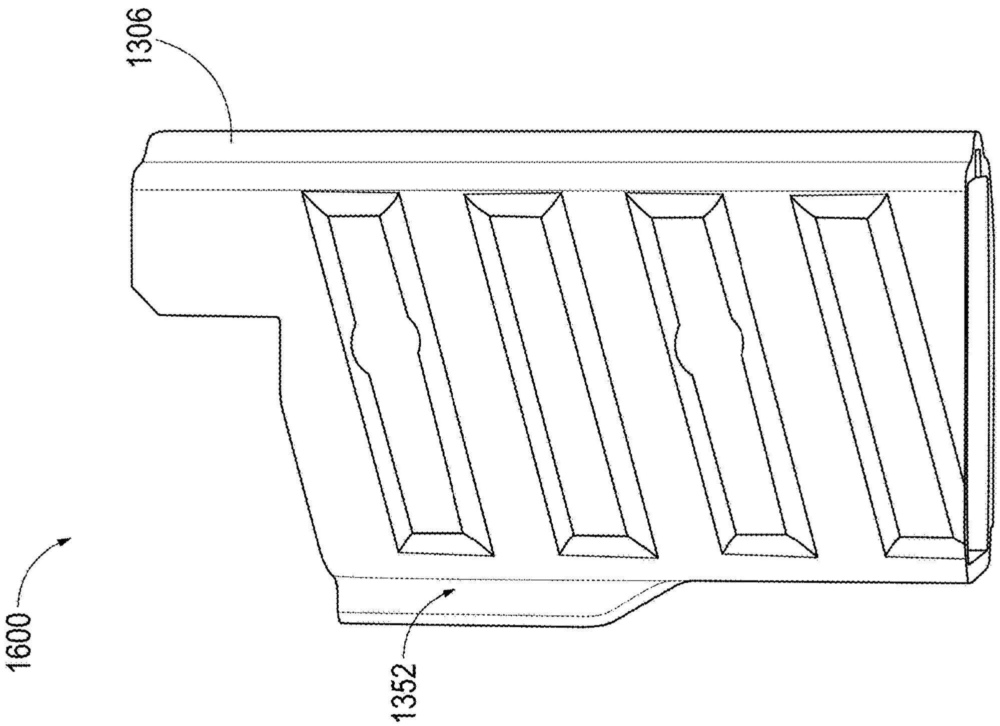

1306
1600
1352

FIG. 16

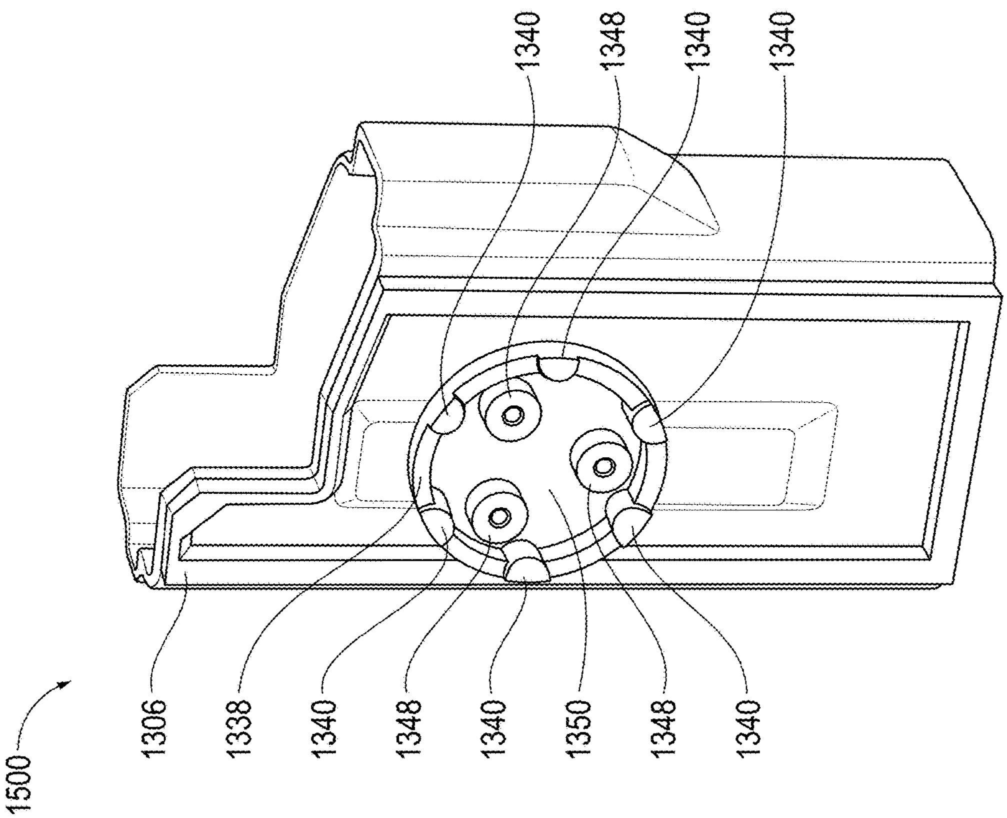

1340
1348
1340
1340
1500
1306
1338
1340
1348
1340
1350
1348
1340

FIG. 15

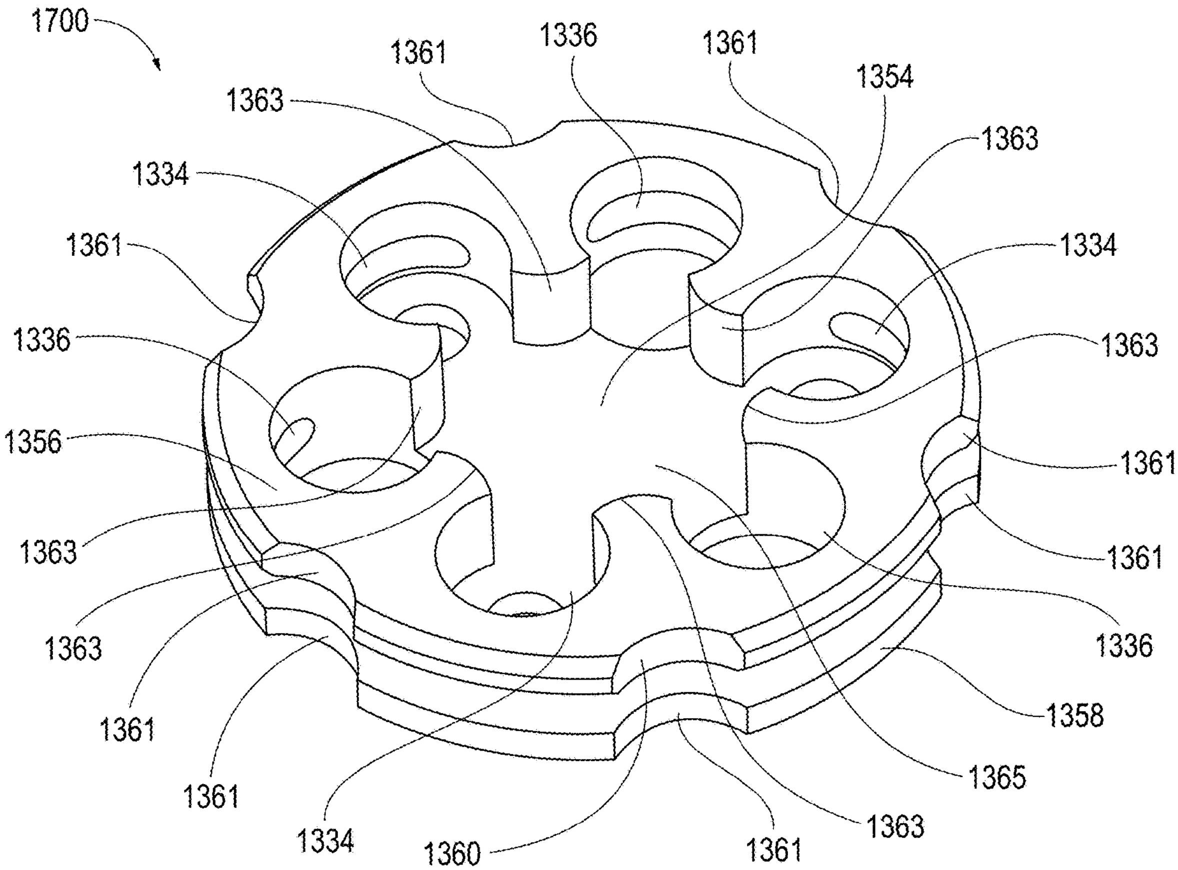

1700
1361
1336
1361
1354
1363
1363
1334
1334
1361
1363
1336
1356
1361
1363
1361
1363
1336
1361
1358
1361
1365
1361
1363
1334
1360
1361

FIG. 17

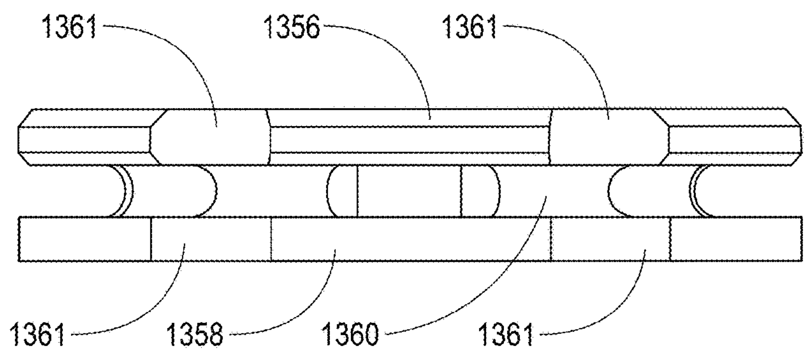

1361
1356
1361
1361
1358
1360
1361

FIG. 18

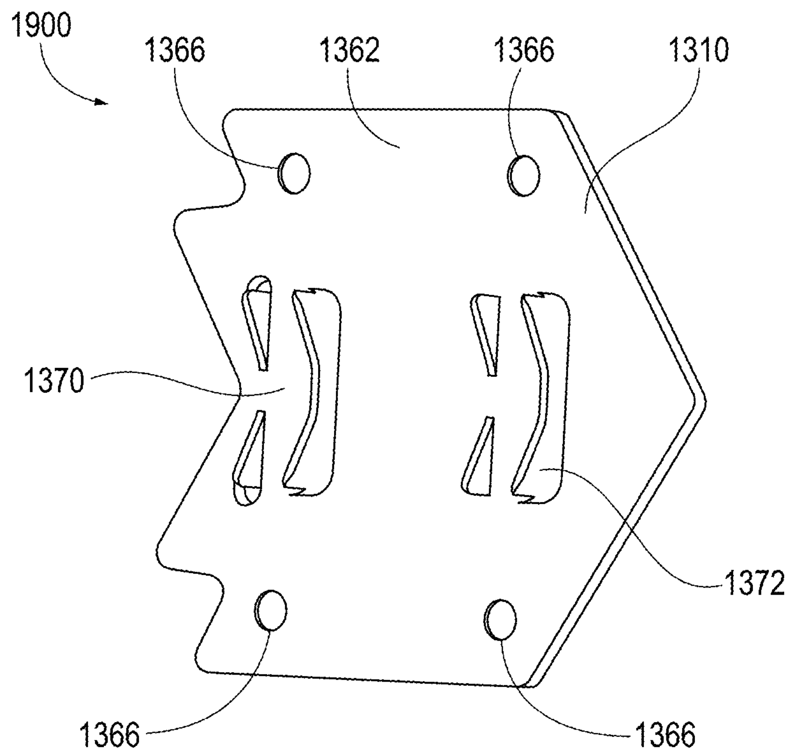

1900
1366
1362
1366
1310
1370
1372
1366
1366

FIG. 19

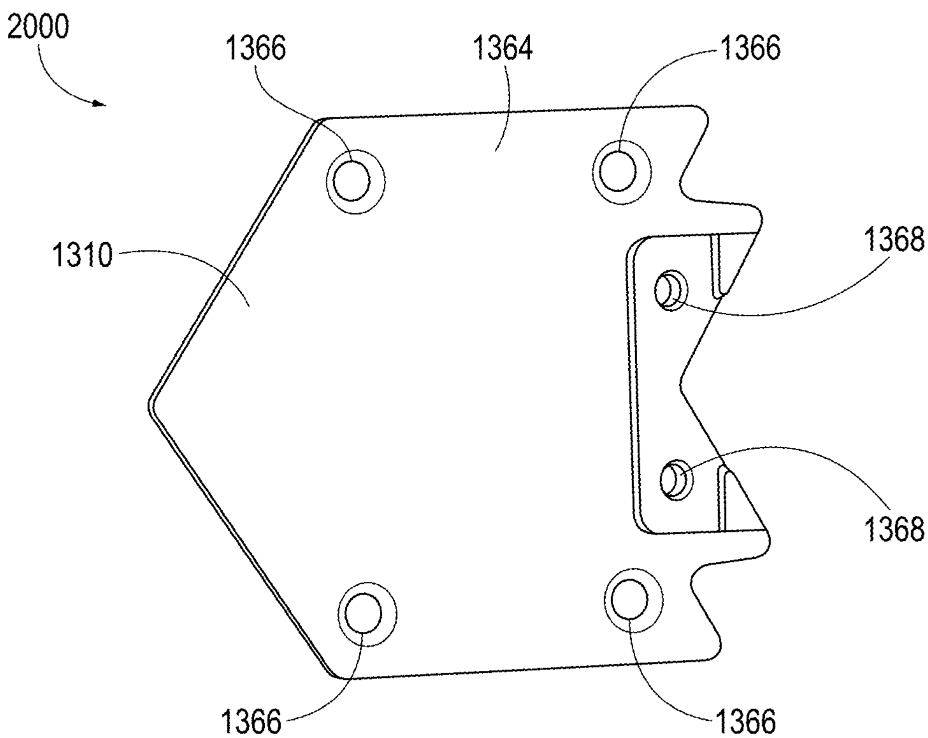

2000
1366
1364
1366
1310
1368
1368
1366
1366

FIG. 20

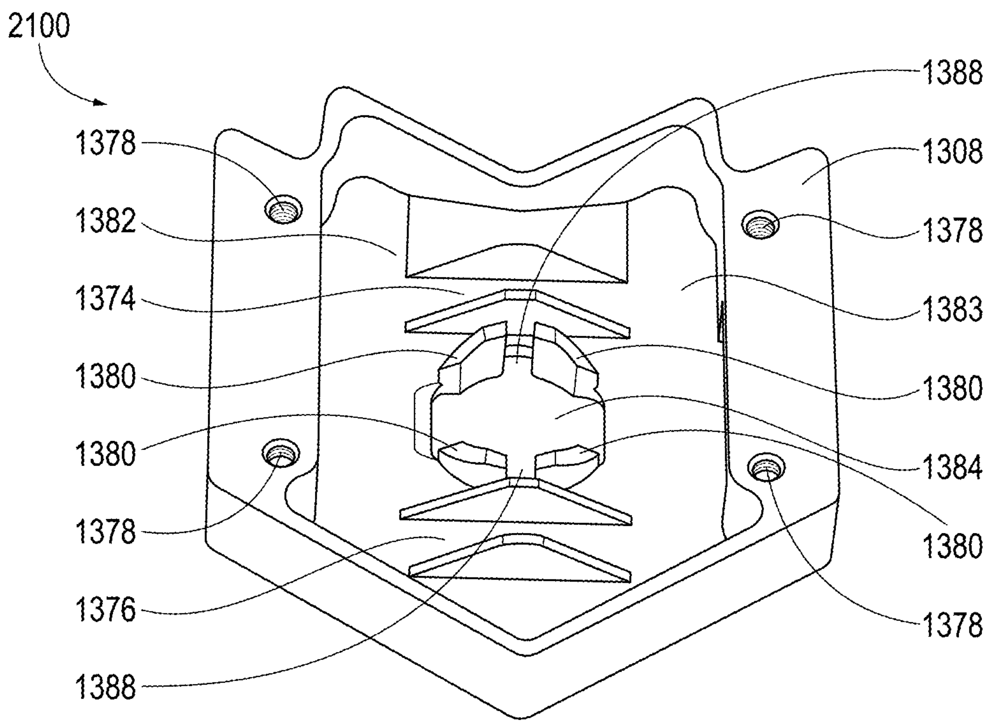

2100
1378
1382
1374
1380
1380
1378
1376
1388
1388
1308
1378
1383
1380
1384
1380
1378

FIG. 21

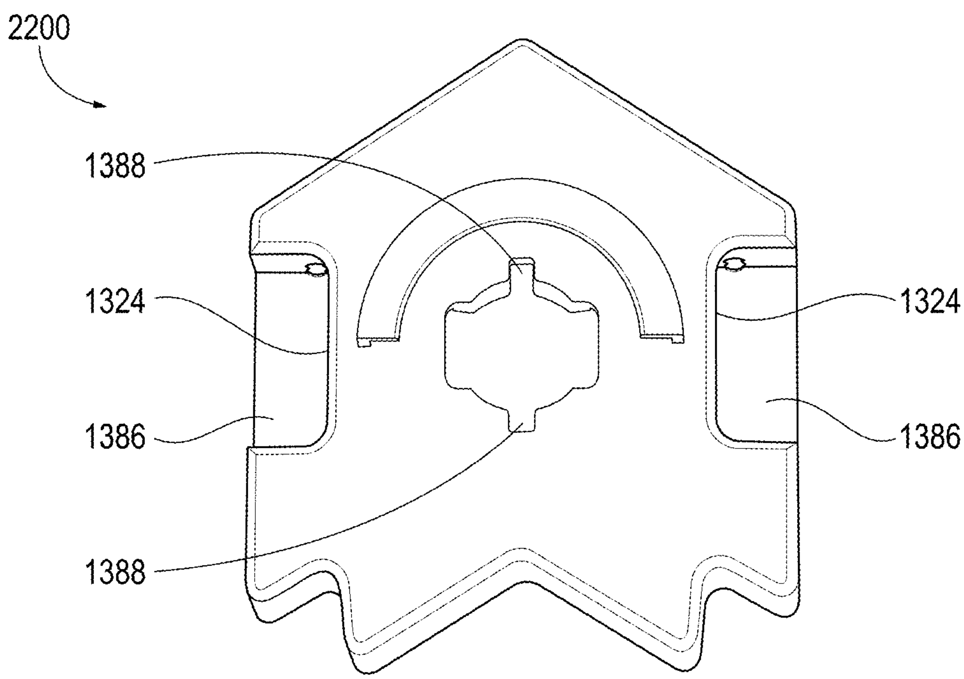

2200
1388
1324
1386
1388
1324
1386

FIG. 22

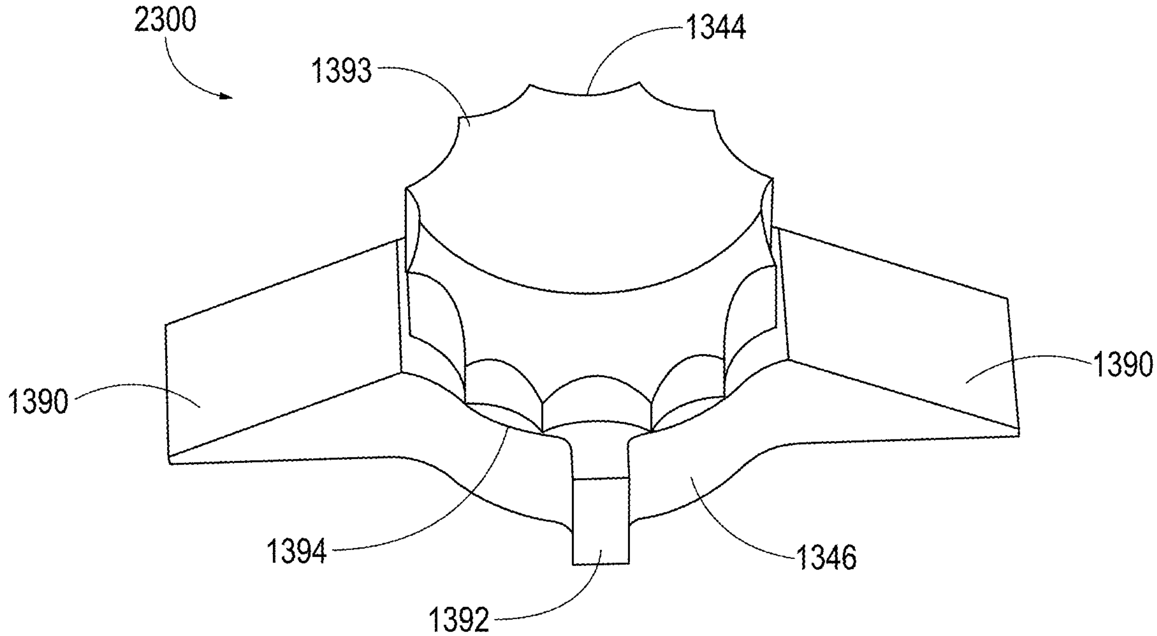

2300
1344
1393
1390
1390
1394
1346
1392

FIG. 23

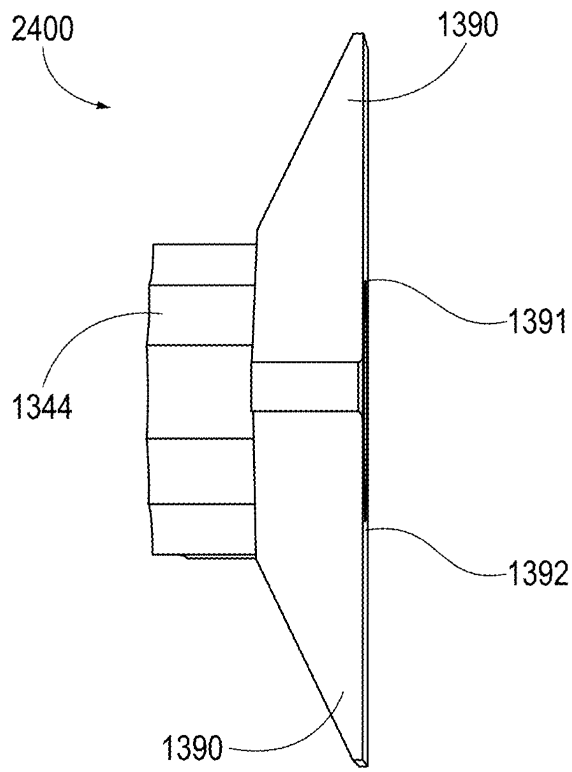

2400
1390
1391
1344
1392
1390

FIG. 24

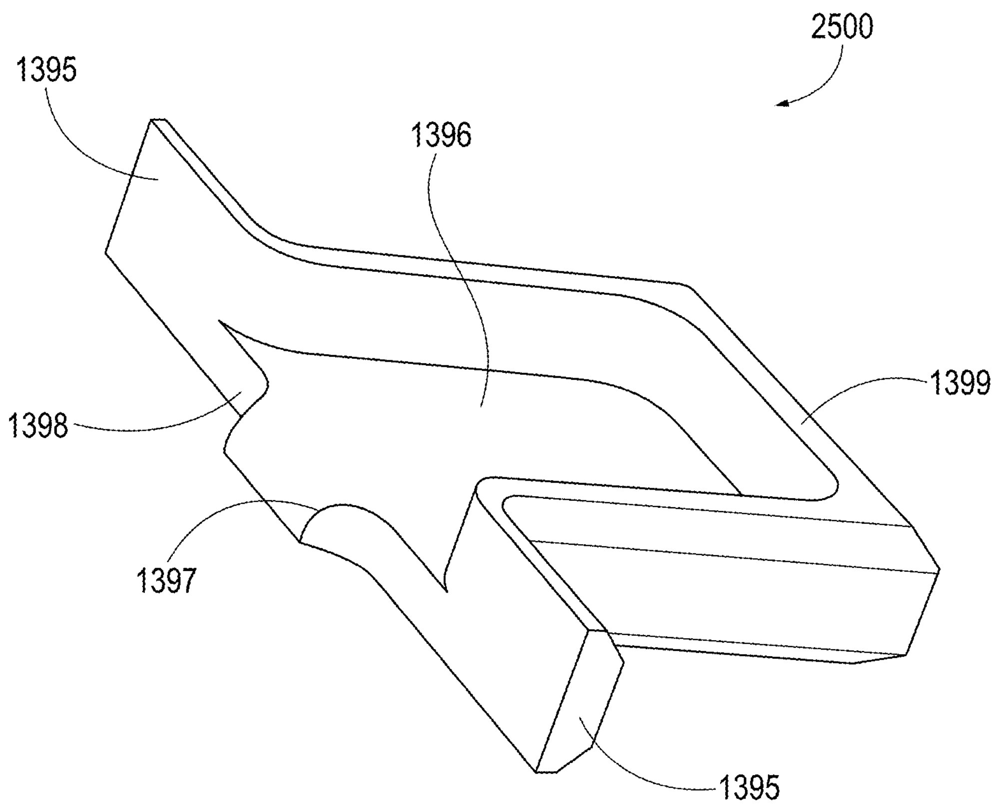


FIG. 25

MODULAR HOLSTER ASSEMBLY

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to U.S. Provisional Application No. 63/262,047, filed Oct. 4, 2021, and entitled "MODULAR HOLSTER," the subject matter of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to holsters for tools, and more specifically, for firearms.

BACKGROUND

A holster is a device used to hold or restrict an undesired movement of a firearm (e.g., a handgun), most commonly in a location where the firearm can be easily withdrawn for use. Holsters are often attached to a belt or waistband, but holsters may be attached to other locations of a body, or at a readily accessible area. Holsters vary in the degree to which such devices secure and/or protect the firearm. For example, law enforcement holsters have a strap over a top of the holster to make the firearm less likely to fall out of the holster and/or harder for another person to retrieve the firearm. Some holsters have a flap over the top to protect the handgun from the elements.

SUMMARY

Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an extensive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

In an example, a mounting apparatus for releasably connecting a holster can include at least one clip and a canal. The canal can include a grooved path that can extend along a base of the canal, and an inner channel that can receive a plunger of a plunger device to secure the holster to mounting apparatus. The mounting apparatus can include a sliding pusher that can disengage the holster from the mounting apparatus.

In another example, a holster system can include a holster for supporting a tool, and a securing apparatus fixed to the holster. The securing apparatus can include a gear and a plunger device. The plunger device can include a plunger. The plunger device can be positioned at least partially through the gear so that the plunger extends away from the gear. The system can further include a mounting apparatus that can include a clip and a canal. The canal can include a grooved path that can extend along a base of the canal, and an inner channel that can receive the plunger to secure the holster to mounting apparatus. The grooved path can gradually narrows toward the inner channel from an opening to the canal.

In an even further example, a holster system can include a clip, and holster for supporting a tool. The holster can include a ring support structure that can have a cavity that can include a ring attachment structure. The ring attachment structure can include retention portions for engaging a gear head of a slide lock. The holster system can further include a mounting apparatus that can include a cavity that includes the slide lock, and an opening through which the gear head of the slide lock can protrude. The mounting apparatus can further include openings through which portions of pinch plates can extend for disengaging the holster from the mounting apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

The following figures are included to illustrate certain aspects of the present disclosure, and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, as will occur to one of ordinary skill in the art and having the benefit of this disclosure.

FIG. **1** is a first perspective view of an example mounting apparatus.

FIG. **2** is a second perspective view of the example mounting apparatus.

FIG. **3** is a first perspective view of an example plate.

FIG. **4** is a second perspective view of the example plate.

FIG. **5** is a perspective view of an example plate cover.

FIG. **6** is a perspective view of an example slide pusher.

FIG. **7** is a perspective view of an example block insert.

FIG. **8** is a first perspective view of an example gear.

FIG. **9** is a second perspective view of the example gear.

FIG. **10** is a first perspective view of an example plunger.

FIG. **11** is a second perspective view of the example plunger.

FIG. **12** is a perspective view of an example tool sleeve.

FIG. **13** is a first perspective view of an example modular holster assembly.

FIG. **14** is a second perspective view of the example modular holster assembly.

FIG. **15** is a first perspective view of an example holster.

FIG. **16** is a second perspective view of the example holster.

FIG. **17** is a first perspective view of an example ring attachment structure.

FIG. **18** is a second perspective view of the example ring attachment structure.

FIG. **19** is a first perspective view of an example back plate.

FIG. **20** is a second perspective view of the example back plate.

FIG. **21** is a first perspective view of an example shell.

FIG. **22** is a second perspective view of the example shell.

FIG. **23** is a first perspective view of an example slide lock.

FIG. **24** is a second perspective view of the example slide lock.

FIG. **25** is a perspective view of an example pinch plate.

DETAILED DESCRIPTION

Examples of the present disclosure will now be described in detail with reference to the accompanying figures. Like elements in the various figures may be denoted by similar reference numerals for consistency. Further, in the following detailed description of embodiments of the present disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the claimed subject matter. However, the embodiments disclosed herein may be practiced without these specific details. Additionally, it will be apparent that the scale of the elements presented in the accompanying figures may vary without departing from the scope of the present disclosure. Shapes and/or dimensions shown in the figures are for example, and other shapes and/or dimensions may be used and remain within the scope of the present disclosure, unless specified otherwise.

Generally, holsters are designed for particular models of firearms or other holstered firearms, resulting in different holsters for each firearm. Moreover, because of features such as retention mechanisms or simply because of the design of the firearm, holster bodies are commonly not interchangeable for left- and right-handed users. Thus, a user of several firearms that desires to holster them, must purchase and situate on a belt or waistband multiple different holsters, making the process costly, time consuming, and inconvenient.

According to the examples herein, a modular holster assembly is provided that includes a mounting apparatus and a securing apparatus. The modular holster assembly can be used to permit a user to readily and easily interchange holsters for different types of tools and situate the holsters on the user. For example, a first holster for a first tool can be adapted to support a first securing apparatus, and a second holster for a second tool can be adapted to support a second securing apparatus. The mounting apparatus can be worn by the user by securing the mounting apparatus to a belt of the user. The first securing apparatus can engage the mounting apparatus to secure the first holster to the mounting apparatus. In some examples, second securing apparatus can engage the mounting apparatus to secure the second holster to the mounting apparatus. Accordingly, the modular holster assembly according to the examples herein allows for the user to have access to different types of tools by swapping corresponding tool holsters configured with a respective securing apparatus, as described herein.

In some examples, a holster is a sleeve and can support different type of tools. The sleeve can be configured to support a securing apparatus such as described herein. The securing apparatus can be configured to engage a mounting apparatus secured to a belt of user to fix the sleeve to mounting apparatus. Because the sleeve can support different types of tools, the user can swap out tools while retaining a same holster for the tools. Accordingly, the modular holster assembly as described herein can enable the user to use a single sleeve for different tools thus reducing a need for the user to wear different holsters for different tools. The securing apparatus of the modular holster assembly can be applied to any sleeve size, allowing the user to carry tools of any size. Any user who wishes to purchase a holster for an additional tool need only purchase an additional sleeve if they already have the modular holster assembly, resulting in a lower total cost for owners of two or more tools.

Moreover, the modular holster assembly can improve a comfort, and utility, and accommodate different user preferences. The modular holster assembly allows for the user to rotate the tool, for example, to a desired angle. The modular holster assembly enables the user to lock a tool at any angle based on the user's preference for carrying and drawing the tool. If a user has additional sleeves/holsters for additional tools, it will also allow for the same feel and sensation for any tool when it comes to application, removal, and drawing the tool. If a user already has multiple sleeves, it makes swapping tools easier, as the modular holster assembly does not need to be removed by the user. Additionally, the modular holster assembly enables the user to mount tools in other areas, such as a car or a cabinet, and allows the user to store a tool or transfer the tool without requiring the tool to be removed from the sleeve. The modular holster assembly can also be adapted for other carrying areas, in addition to waist/belt carrying.

Accordingly, the modular holster assembly as described herein enables users of multiple sizes of tools to swap one tool for another quickly and easily, without sacrificing a feel and placement. This allows a user to transition from a full-sized firearm to a compact, a semiautomatic to a revolver, or simply an on-duty caliber to a different caliber for personal carry after a work shift, and all in a more cost- and time-efficient manner than owning and changing entire holsters. As such, the modular holster assembly allows for an easy, efficient, and cost-effective manner of changing firearms for open and concealed carry.

The term "tool" as used herein can include a firearm, a knife, a flashlight, or any object that can extend an individual's or users ability to modify features of a surrounding environment. The term "holster" and its derivatives as used herein can include any sleeve, device and/or apparatus that can be used to hold and/or restrict a movement of a tool. While examples are presented herein wherein the holster is situated on a user, in other examples, the holster may be situated on an object (e.g., under a steering column).

FIGS. **1-2** are first and second perspective views **100** and **200** of an example mounting apparatus **102**. The mounting apparatus **102** includes a mount **104** having sides **106** and **108**. The mount **104** includes a number of channels **110** that extend through the mount **104** and open into the sides **106** and **108** to form respective openings. The mounting apparatus **102** can further include clips **112** that extend along a length of the side **108**. While the example of FIG. **2** illustrates two (2) clips in other examples, a single loop can be used, and positioned along a center length of the side **108**. In some examples, the mount **104** and the clips **112** are composed of different or similar materials. As an example, the materials may include flexible and/or durable materials. In additional or alternative examples, the mount **104** and the clips **112** can be composed of a plastic material (e.g., a thermoplastic material) or metal material. The clips **112** can be used to mount or secure the mounting apparatus **102** to a user (e.g., via a garment, belt, strap, or accessory), or an object (e.g., a surface, furniture, vehicle, locker, etc.).

By way of further example, channels **114** can extend through the clips **112** to respective loop sides to form respective openings. The openings of the clips **112** can be aligned with one of the respective openings of the mount **104** and secured via a fastener (e.g., a screw, a bolt, etc.) to fix the clips **112** to the mount **104** to form respective loops (e.g., for receiving a belt to secure the mounting apparatus **102** to the user). In other examples, a different mechanism can be used to secure the clips **112** to the mount **104**. For example, the clips **112** can be welded to the mount **104** (e.g., using a welding material).

The mount **104** further includes a recess **116** on the side **108**. Channels **118** can extend through the mount **104** to the sides **106** and **108** to form respective openings. The securing of the clips **112** to the mount **104** can form a passageway(s) or loop(s) **120**. In some examples, to secure the mount **104** to a user, a belt of the user can be threaded along a width of the mount **104** through the loop **120**. In further examples, the mount **104** includes a base plate **122** and a base plate cover **124**. Channels **128** can extend through the base plate **122** and the base plate cover **124** to form respective openings. The respective openings of the base plate **122** can be aligned with one of the respective openings of the channels **118** on the side **106** and secured via a fastener (not shown) to fix the base plate **122** to the mount **104**. In other examples, a different mechanism can be used to secure the base plate **122** to the mount **104** such as described herein.

In some example, the base plate **122** can be at least partially hollow to form or support a canal **130** with an opening **132**. As described herein, a securing apparatus on a sleeve can be positioned in the canal **130** to secure the sleeve to the mount **104**. For example, the canal **130** can include a grooved path **134**. The grooved path **134** can extend along a base **138** of the canal **130** from the opening **132** toward an inner channel **136** that opens to the canal **130**. In some examples, the base plate **122** can include a side opening **140** through which a sliding pusher **142** can be positioned. The sliding pusher **142** can extend partially through the side opening **140** toward the inner channel **136**, as shown in FIG. **1**.

FIGS. **3-4** are first and second perspective views **300** and **400** of the base plate **122**, as shown in FIGS. **1-2**. Thus, reference can be made to example of FIGS. **1-2** in the example of FIGS. **3-4**. As shown in FIG. **3**, the base plate **122** includes the grooved path **134**. The grooved path **134** can have a concave shape such that a width **144** of the grooved path **134** gradually narrows as the grooved path **134** approaches the inner channel **136**. The grooved path **134** includes a number of concave steps or grooves **146** that are formed along the base **138** of the canal **130**. As the width of the grooved path **134** gradually narrows toward the inner channel **136** each concave step **146** can have a curvature that changes (e.g., increases) with respect to a neighboring concave step **146** toward the inner channel **136**. In further examples, the canal **130** of the base plate **122** has a concave end **148** that can be opposite to an opening end of the canal (corresponding to the opening **132**), as shown in FIG. **3**. While the example of FIG. **3** illustrates the canal **130** as having the concave end **148** in other examples, the canal **130** have a different shape (e.g., a rectangular end). The canal **130** can have a depth so that a gear of the securing apparatus can be inserted and guided toward the inner channel **136**.

The concave end **148** can include teeth **150**. While the example of FIG. **3** illustrates a given number of teeth, in other examples, the concave end **148** can include a different number of teeth. By way of further example, the inner channel **136** includes a circular portion **152** and rectangular openings **154** and **156** that are formed in the base **138** of the base plate **122**. Complementary circular and rectangular openings are also formed on a side **159** of the base plate **122** opposite a side **157** on which the circular and rectangular portions **152**, **154**, and **156** are formed. Additionally, the side **159** of the base plate **122** include respective cavities **160** and **162**, and a cavity coupling channel **164** for connecting the cavities **160** and **162**. The cavity **160** includes the side opening **140**. The sliding pusher **142** can be extended toward the circular portion **152** through the cavity **160** into the cavity coupling channel **164**. In some examples, the cavity **162** includes a side opening **166**, and the sliding pusher **142** can be extended toward the circular portion **152** through the cavity **162** into the cavity coupling channel **164**. Through which of the side openings **140** or **166** that the sliding pusher **142** can be extended can be based on which side of the user the mounting apparatus **102** is worn. As shown in FIGS. **3-4**, the base plate **122** is a given width and thus include channels **168** that can extend through the base plate **122** to form respective openings on the sides **157** and **159** of the base plate **122**.

FIG. **5** is a perspective view of the base plate cover **124**, as shown in FIG. **1**. Thus, reference can be made to example of FIGS. **1-4** in the example of FIG. **5**. As shown in FIG. **5**, the base plate cover **124** is of a given width and include channels **170** that extend through the base plate cover **124** to form respective openings on sides **171** and **172** of the base plate cover **124**. In some examples, the channels **168** and **170** can define or form the channel **128**, as shown in FIG. **1**. The base plate cover **124** can include an opening **174** for allowing the gear of the securing apparatus to be inserted and guided toward the inner channel **136**. Once the respective openings of the base plate **122** are aligned with the respective openings of the base plate cover **124**, the base plate **122** and the base plate cover **124** can be secured (e.g., via fasteners, or some different securing mechanism).

FIG. **6** is a perspective view of the slide pusher **142**, as shown in FIG. **1**. Thus, reference can be made to example of FIGS. **1-5** in the example of FIG. **6**. The slide pusher **142** includes a body **176**, wings **178** that extend from the body **176**, a tail end **180**, and a head end **182**. The wings **178** of the slide pusher **142** can be positioned within one of the cavities **160** and **162** so that the tail end **180** of the slide pusher **142** extends away from the corresponding cavity **160** and **162** and through one of the side openings **140** and **166**. The wings **178** can be used to retain the slide pusher **142** within one of the cavities **160** and **162**. Once positioned, the head end **182** of the slide pusher **142** can extend into the cavity coupling channel **164**. The head end **182** of the slide pusher **142** can include a sloped portion **184** to enable manual release or disengagement of the sleeve from the mounting apparatus **102**, as described herein.

FIG. **7** is a perspective view **700** of a block insert **186** that can be used in the mounting apparatus **102**, as shown in FIG. **1**. Thus, reference can be made to example of FIGS. **1-6** in the example of FIG. **7**. The block insert **186** can be inserted into a respective one of the cavities **160** and **162**, as shown in FIG. **2**. For example, if the wings **178** of the slide pusher **142** are positioned within the cavity **160**, the block insert **186** can be positioned within the cavity **162**. The block insert **186** can be used to stop the slide pusher **142** such as during manual release or disengagement.

FIGS. **8-9** are first and second perspective view of a gear **188** that can be used for securing the sleeve to the mounting apparatus **102**, as shown in FIG. **1**. Thus, reference can be made to example of FIGS. **1-7** in the example of FIGS. **8-9**. The gear **188** includes a number of teeth **190** that protrude axially and circumferentially from a center of the gear **188**. The gear **188** can have a first face **192** and a second face **193**. A tube body **194** can extend away from the second face **193** of the gear **188** and be of sufficient length to support a plunger device, such as described herein.

FIGS. **10-11** are first and second perspective views **1000** and **1100** of a plunger device **195** that can be used during securing of the sleeve to the mounting apparatus **102**, as shown in FIG. **1**. Thus, reference can be made to example of FIGS. **1-9** in the example of FIGS. **10-11**. The plunger device **195** can include a plunger **196** and a plunger body **197**. The plunger device **195** can be positioned at least partially through the gear **188** so that the plunger **196** extends away from the first face **192** of the gear **188**. The plunger body **197** can extend at least partially through the tube body **194**. The plunger body **197** includes a spring **198** that can be compressed during securing of the sleeve to the mounting apparatus **102**. The spring **198** can be coupled to the plunger **196**. The spring **198** can keep a constant pressure on the plunger **196** when the plunger **196** is in a resting position. In some examples, a force can be applied on the plunger **196**, such as during the securing of the sleeve to the mounting apparatus **102**, and the plunger **196** can be extended. The plunger **196** can be pushed back into the plunger body **197** when the force on the plunger **196** exceeds a spring force being applied by the spring **198**.

FIG. **12** is a perspective view **1200** of an example sleeve **202** with a securing apparatus **204**. The securing apparatus **204** includes the gear **188** and the plunger device **195** positioned therein. The plunger **196** of the plunger device **195** can protrude from the gear **188**. Thus, reference can be made to the examples of FIGS. **1-11** in the example of FIG. **12**. The sleeve **202** can be used to retain or hold a tool, for example, a firearm. For example, to secure the sleeve **202** to the mounting apparatus **102**, the sleeve **202** is positioned with respect to the mounting apparatus **102** so that the securing apparatus **204** is moved toward the opening **132** of the canal **130** of the base plate **122**. As the plunger **196** of the securing apparatus **200** enters the opening **132**, the plunger **196** can engage the grooved path **134** of the base **138** of the base plate **122**. The grooved path **134** allows for the plunger **196** to clear the canal **130** toward the inner channel **136** as the sleeve **202** is pushed toward (e.g., downward) the inner channel **136**. Each concave step **146** gradually depresses the plunger **196** inward (e.g., into the plunger body **197**) as the plunger **196** is being pushed toward the inner channel **136**. Once the plunger **196** reaches a final or last concave step **146**, the plunger **196** can become flush with an outward surface of the first face **192** of the gear **188**. The plunger **196** can spring back into place through the inner channel **136** toward the side **159** of the base plate **122** in response to the plunger **196** reaching the inner channel **136** in the canal **130**.

Additionally, the teeth **150** contained within the canal **130** can interlock with the teeth **190** of the gear **188**, for example, once the plunger **196** reaches the inner channel **136** or springs backs into place into the inner channel **136**. Because the teeth **150** and **190** are interlocked, the sleeve **202** and the mounting apparatus **102** can be connected so that the sleeve **202** does not rotate with respect to the mounting apparatus **102**. In some examples, the sliding pusher **142** is omitted from the mounting apparatus **102** and the mounting apparatus **102** includes a push button. The push button can be a sliding piece underneath and above a recess that protrudes out of a top of a canal piece. To remove or disconnect the sleeve **202** from the mounting apparatus **102**, a pressure (e.g., downward pressure) can be applied to the push button on the canal **130**. The pressure can disengage the gear **188** from the canal **130** to move push button under the plunger **196**, pushing the plunger **196** up to be flush with a floor (e.g., a surface of the base **138**) of the canal **130** so that the sleeve **202** can be separated from the mounting apparatus **102**. In some examples, the sliding pusher **142** can be used for manual release or disengagement of the sleeve **202** with the mounting apparatus **102**. For example, a pressure can be applied to the sliding pusher **142** to move or push the wings **178** of the sliding pusher **142** along a respective cavity **160** and **162** until the sloped portion **184** of the sliding pusher **142** is moved under the plunger **196** to push the plunger **196** up to be flush with the floor of the canal **130**. The sleeve **202** can be separated from the mounting apparatus **102** in response to being flush with the floor.

FIG. **13** is a perspective view **1300** of an example of a holster assembly **1302**. The holster assembly **1302** includes a mounting apparatus **1304** and a holster **1306**. The holster **1306** can have an ergonomic shape and support a firearm, such as a handgun (not shown). While the example of FIG. **13** illustrates the holster **1306** as supporting a handgun in other examples, the holster **1306** can be configured to support a different type of firearm. Moreover, while the example of FIG. **13** illustrates a holster for a firearm, in other examples, the holster **1306** can be designed to support a different tool (e.g., a knife, a flashlight, etc.). The mounting apparatus **1304** includes a shell **1308** and a back plate **1310**. A number of channels **1312** can extend through the shell **1308** and the back plate **1310** into openings on respective sides **1314** and **1316** of one of the shell **1308** and the back plate **1310**. As shown in FIG. **13**, a fastener can be positioned through the channels **1312** to couple the shell **1308** and the back plate **1310**. In other examples, a different mechanism can be used to secure the shell **1308** and the back plate **1310**, such as described herein.

The holster assembly **1302** can further include clips **1318** that extend along a length of the side **1314**. While the example of FIG. **13** illustrates two (2) clips in other examples, a single clip can be used, and positioned along a center length of the side **1314**. In some examples, the mounting apparatus **1304**, the holster **1306** and/or the clips **1318** are composed of different or similar materials. As an example, the materials may include flexible and/or durable materials. In additional or alternative examples the mounting apparatus **1304**, the holster **1306** and/or the clips **1318** can be composed of a plastic material (e.g., a thermoplastic material) or metal material. The clips **1318** can be used to mount or secure the mounting apparatus **1304** and thus the holster **1306** to a user (e.g., via a garment, belt, strap, or accessory), or an object (e.g., a surface, furniture, vehicle, locker, etc.).

In some examples, the clips **1318** include a number of openings **1320** for use in securing the clips **1318** to the back plate **1310**. In some examples, the back plate **1310** includes openings for securing the clips **1318** to the shell **1308** of the mounting apparatus **1304**. For example, the openings **1320** of the clips **1318** can be aligned with one of the openings of the back plate **1310** and secured via a fastener (e.g., a screw, as shown in FIG. **13**) to fix the clips **1318** with respect to the mounting apparatus **1304**. In other examples, a different mechanism can be used to secure the clips **1318** to the back plate **1310**, as described herein. By way of further example, the shell **1308** can further include openings **1324** through which a pinch plate **1326** can be positioned. The pinch plate **1326** can extend partially through one of the openings **1324**, as shown in FIG. **13**. The securing of the clips **1318** to the shell **1308** of the mounting apparatus **1304** can form a passageway(s) or loop(s) **1328**. In some examples, to secure the mounting apparatus **1304** to a user, a belt of the user can be threaded along a width of the mounting apparatus **1304** through the loop(s) **1328**.

FIG. **14** is a perspective view **1400** of an example of the holster assembly **1302**, as shown in FIG. **13**. Thus, reference can be made to the example of FIG. **13** in the examples of FIG. **14**. In the example of FIG. **14**, the holster assembly **1302** is disassembled, such that the holster **1306** is disengaged from the mounting apparatus **1304**. As shown in FIG. **14**, the holster **1306** includes a surface **1330** with a ring support structure **1332** located thereon. In some examples, the holster **1306** can be molded so that the ring support structure **1332** is part of the holster **1306**. The ring support structure **1332** includes a first set of channels **1334** and a second set of channels **1336**. The first set of channels **1334** can be used for securing a ring attachment structure **1338** to the ring support structure **1332**. As shown in FIG. **14**, a respective fastener (e.g., a mating bolt) can be positioned through the channels **1334** to couple the ring attachment structure **1338** to the ring support structure **1332**. In other examples, a different mechanism can be used to secure ring attachment structure **1338** to the ring support structure **1332**. The ring support structure **1332** can include ribs **1340** that can protrude toward a center of the ring support structure **1332**. In some examples, the ribs **1340** are curved and can have a perimeter that is similar to a semicircle. The ribs **1340** can function as guides during placement of the ring attachment structure **1338** at least partially within the ring support structure **1332**. By way of further example, the mounting apparatus **1304** can include a central opening **1342** through which a gear head **1344** of a slide lock **1346** (not shown in FIG. **13**) can protrude.

FIGS. **15-16** are first and second perspective view of the holster **1306**, as show in FIGS. **13-14**. Thus, reference can be made to the examples of FIGS. **13-14** in the examples of FIGS. **15-16**. In some examples, disposed within the ring support structure **1332** are a number of securing portions **1348**. In some examples, the securing portions **1348** are fasteners that resemble a nut and thus can be threaded with the mating bolt. During securing of the ring attachment structure **1338** to the ring support structure **1332**, the ring attachment structure **1338** can be aligned with respect to the ring support structure **1332** so that each channel of the first set of channels **1334** is aligned with one of the securing portions **1348**. The ring support structure **1332** can have a greater circumference than the ring attachment structure **1338**. Thus, the ring attachment structure **1338** can at least be partially positioned within a cavity **1350** of the ring support structure **1332**. Once the ring attachment structure **1338** is partially located within the cavity **1350**, the respective fastener can be positioned through each of the first set of channels **1334** and mated with one of the securing portions **1348**. In some examples, the holster **1306** can include ergonomic grooves **1352** to provide a ribbed texture for increasing a use of the holster **1306** on an opposite side of the ring support structure **1332**.

FIGS. **17-18** are first and second perspective views of the ring attachment structure **1338**, as shown in FIG. **14**. Thus, reference can be made to the examples of FIGS. **13-16** in the examples of FIGS. **17-18**. The ring attachment structure **1338** includes a base **1354**. The first and second channels **1334** and **1336** can open into the base **1354**. In some examples, the ring attachment structure **1338** includes first and second circular portions **1356** and **1358**. The first and second circular portions **1356** and **1358** can be coupled via a coupling portion **1360**. Each of the first and second circular portions **1356** and **1358** can include indents **1361**. During securing of the ring attachment structure **1338** to the ring support structure **1332**, the indents **1361** can be configured to receive one of the ribs **1340**. Thus, the ribs **1340** can be retained with one of the indents **1361** to prevent rotational motion of the ring attachment structure **1338** about its axis. By way of further example, the coupling portion **1360** includes retention portions **1363**. The retention portions **1363** can be used to retain the gear head **1344**. For example, the gear head **1344** can be extended into a cavity **1365** toward the base **1354** to come into contact with the retention portions **1363**. The retention portions **1363** can be used for fixing the gear head **1344** to the mounting apparatus **1304** to the holster **1306**.

FIGS. **19-20** are first and second perspective views **1900** and **2000** of back plate **1310**, as shown in FIGS. **13-14**. Thus, reference can be made to the examples of FIGS. **13-18** in the examples of FIGS. **19-20**. The back plate **1310** has sides **1362** and **1364**. In some examples, the side **1364** can correspond to the side **1314**, as shown in FIGS. **13-14**. The back plate **1310** includes first openings **1366** and a second openings **1368**. During assembly of the mounting apparatus **1304**, the shell **1308** can be coupled to the back plate **1310** such that the first openings **1366** of the back plate **1310** are aligned with openings of the shell **1308**. Thus, the channel **1312** can extend through the shell **1308** and the back plate **1310** into corresponding first openings of the shell **1308** and the back plate **1310**. During assembly of the mounting apparatus **1304** and the clips **1318**, the clips **1318** can be coupled to the back plate **1310** such that the second openings **1368** are aligned with the openings **1320** of the clips **1318**. By way of further example, the back plate **1310** includes plate interlocking structures **1370** and **1372**. The plate interlocking structures **1370** and **1372** can be configured to engage corresponding shell interlocking structures **1374** and **1376** of the shell **1308** that can protrude in a cavity **1383** of the shell **1308**, as shown in FIG. **21**

FIGS. **21-22** are first and second perspective views **2100** and **2200** of shell **1308**, as shown in FIGS. **13-14**. Thus, reference can be made to the examples of FIGS. **13-20** in the examples of FIGS. **21-22**. The shell **1308** includes channels **1378** that can be aligned with one of the first openings **1366** during assembly of the mounting apparatus **1304**. The shell **1308** can further include lips **1380** that can protrude from a base **1382** of the shell **1308** and thus an opening **1384** of the shell **1308**. In some examples, the opening **1384** can correspond to the central opening **1342**, as shown in FIG. **14**. The lips **1380** can be used to retain the gear head **1344** of the slide lock **1346**, as shown in FIG. **14**, for example, during assembly of the mounting apparatus **102**. By way of further example, the shell **1308** includes recesses **1386**. The openings **1324** of the shell **1308** can open into recesses **1386**. A respective recess **1386** can be used to retain a respective leg **1390** of the slide lock **1346**, as shown in FIG. **22**, during assembly of the mounting apparatus **1304**. In further examples, the opening **1384** of the shell **1308** can include notches **1388**. The notches **1388** can receive a respective lip **1392** of the slide lock **1346**, as shown in FIG. **22**, during assembly of the mounting apparatus **1304**.

FIGS. **23-24** are first and second perspective views **2300** and **2000** of the slide lock **1346**, as shown in FIGS. **13-14**. Thus, reference can be made to the examples of FIGS. **13-22** in the examples of FIGS. **22-23**. The slide lock **1346** can include a base **1394** on which the gear head **1344** can be situated. The legs **1390** of the slide lock **1346** can extend away from the base **1394** in opposite directions, as shown in FIGS. **22-23**. In some examples, the gear head **1344** includes a number of ribs **1393** that form an outer portion (e.g., an outer shell) of the gear head **1344**, such that the gear head **1344** resembles a gear. During assembly of the mounting apparatus **1304**, the slide lock **1346** can be positioned within the cavity **1383** so that a bottom face **1391** lays on a surface of the base **1382** of the shell **1308**. The legs **1390** of the slide lock **1346** can extend toward one of the openings **1324** of the shell **1308**. The legs **1390** can extend into a cavity **1396** of a corresponding pinch plate **1326**, such as shown in FIG. **24**

FIG. **25** is a perspective view **2500** of one of the respective pinch plates **1326**, as shown in FIGS. **13-14**. Thus, reference can be made to the examples of FIGS. **13-24** in the example of FIG. **25**. The pinch plate **1326** includes side protruding flanges **1395** that extend away from a body of the pinch plate **1326**. The pinch plate **1326** further includes the cavity **1396** and a forward facing flange **1397** that extends away from a face portion **1398** of the pinch plate **1326**. During assembly of the mounting apparatus **1304**, the side protruding flanges **1395** are positioned within the cavity **1383** of the shell **1308** such that a rear portion **1399** of the pinch plate **1326** can extend into the recess **1386**. The side protruding flanges **1395** can move within the cavity **1383** (e.g., in response to a pressure) so that the forward facing flange **1397** can engage one of the legs **1390** of the slide lock **1346**.

For example, to secure the holster **1306** to the mounting apparatus **1304**, the holster **1306** can be moved toward the mounting apparatus **102** so that the ring attachment structure **1338** (secured to the ring support structure **1332**) is aligned to receive the gear head **1344** of the mounting apparatus **102**. The gear head **1344** can be oriented toward the surface of the base **1354**. As the gear head **1344** enters the cavity **1365** of the ring attachment structure **1338**, the gear head **1344** engages (e.g., slides against) the retention portions **1363** and the retention portions **1363** create a force to secure the gear head **1344** therein and thus secure the mounting apparatus to the holster **1306**. In some examples, the forward facing flange **1397** of each pinch plate **1326** can be adjacent to the opening **1384** (e.g., located underneath). The gear head **1344** can push the forward facing flanges **1397** away from the opening **1384** as the gear head **1344** is pushed toward the base **1354**. Radial movement of the holster **1306** can be minimized or prevented because at least some of the ribs **1393** of the gear head **1344** are located between neighboring retention portions of the retention portions **1363**.

To disengage the holster **1306** from the mounting apparatus **1304**, a force (e.g., pressure, or pulling force) can be applied by the user to the holster **1306** to overcome the force that is securing the gear head **1344** in the cavity **1365** and thus separate the holster **1306** from the mounting apparatus **1304**. For example, the user can apply a respective force to each pinch plate **1326** to move the pinch plate **1326** inward toward the central opening **1342**. The forward facing flange **1397** of each pinch plate **1326** can slide under the gear head **1344** to push the gear head **1344** away from the base **1354** of the ring attachment structure **1338**. This can overcome the force that was securing the gear head **1344** within the cavity **1365** to free the holster **1306** from the mounting apparatus **1304**.

As used herein, the term "coupled" or "coupled to" or "connected" or "connected to" or "attached" or "attached to" may indicate establishing either a direct or indirect connection, and is not limited to either unless expressly referenced as such. Wherever possible, like or identical reference numerals are used in the figures to identify common or the same elements. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale for purposes of clarification.

It is to be further understood that like or similar numerals in the drawings represent like or similar elements through the several figures, and that not all components or steps described and illustrated with reference to the figures are required for all embodiments or arrangements.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "contains", "containing", "includes", "including," "comprises", and/or "comprising," and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Terms of orientation are used herein merely for purposes of convention and referencing and are not to be construed as limiting. However, it is recognized these terms could be used with reference to an operator or user. Accordingly, no limitations are implied or to be inferred. In addition, the use of ordinal numbers (e.g., first, second, third) is for distinction and not counting. For example, the use of "third" does not imply there is a corresponding "first" or "second." Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including," "comprising," "having," "containing," "involving," and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

While the disclosure has described several exemplary embodiments, it will be understood by those skilled in the art that various changes can be made, and equivalents can be substituted for elements thereof, without departing from the spirit and scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation, or material to embodiments of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiments disclosed, or to the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims.

The invention claimed is:

1. A holster system comprising:

a clip;

a holster for supporting a tool, the holster comprising a ring support structure having a cavity comprising a ring attachment structure, the ring attachment structure comprising retention portions for engaging a gear head of a slide lock; and

a mounting apparatus comprising a cavity that includes the slide lock, and an opening through which the gear head of the slide lock protrudes, the mounting apparatus further comprising openings through which portions of pinch plates can extend for disengaging the holster from the mounting apparatus.

2. The holster system of claim **1**, wherein each pinch plate comprises a cavity, and the slide lock includes legs on opposite sides, wherein the cavity of each pinch plate receives a respective leg of the legs of the slide lock.

3. The holster system of claim **2**, wherein each pinch plate comprises a forward facing flange for sliding under the gear head to push the gear head away from a base of the ring attachment structure in response to the pinch plates being engaged.

4. The holster system of claim **3**, wherein the gear head comprises ribs, and at least some of the ribs are located between neighboring retention portions of the retention portions to prevent a rotation of the holster with respect to the mounting apparatus.

5. The holster system of claim **4**, wherein the opening comprises notches and the slide lock comprises lips, the lips being located within a respective notch of the notches to prevent a rotation of the slide lock.

* * * * *